University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Bias and Errors

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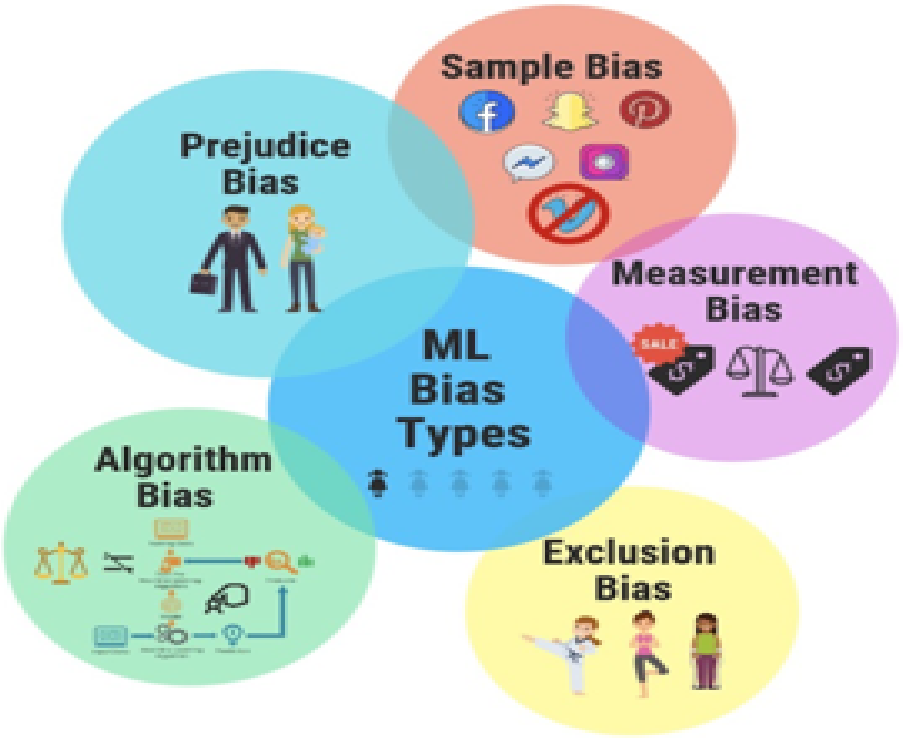
Agenda

1. Introduction
2. Types of Bias
3. Examples

Introduction

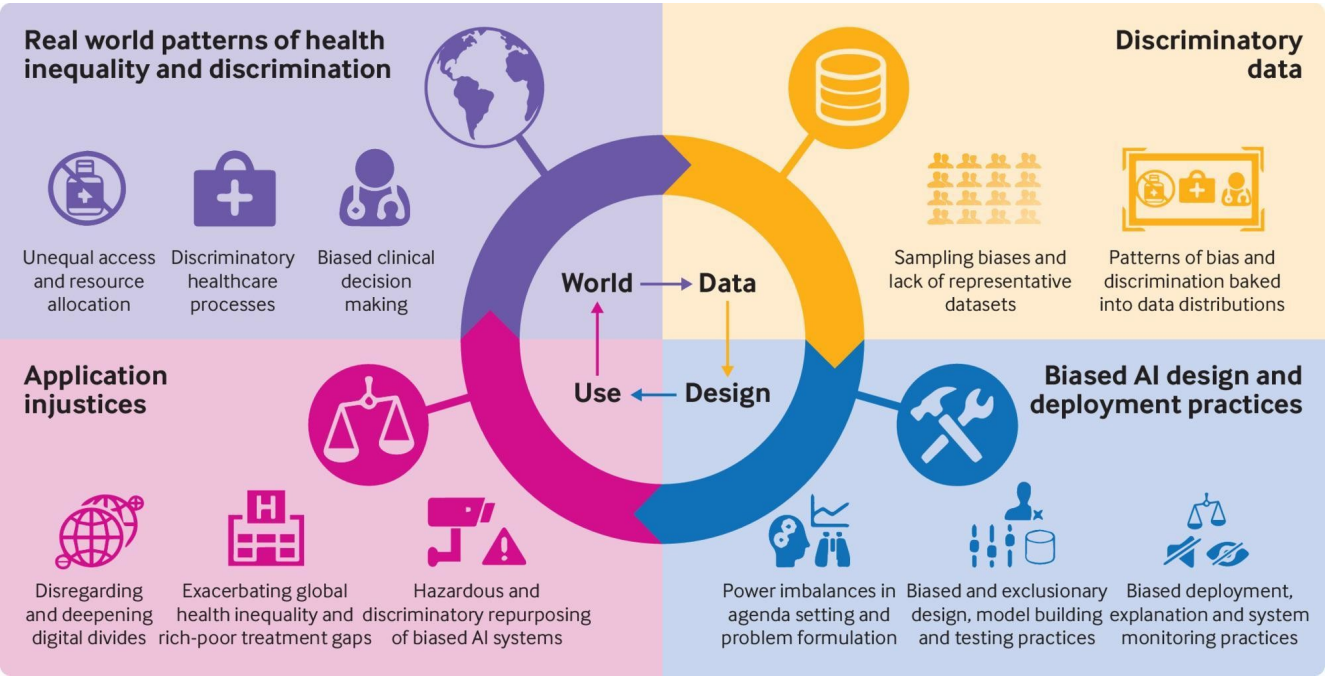


Introduction

"Man was created at the end of the week's work, when God was already tired."

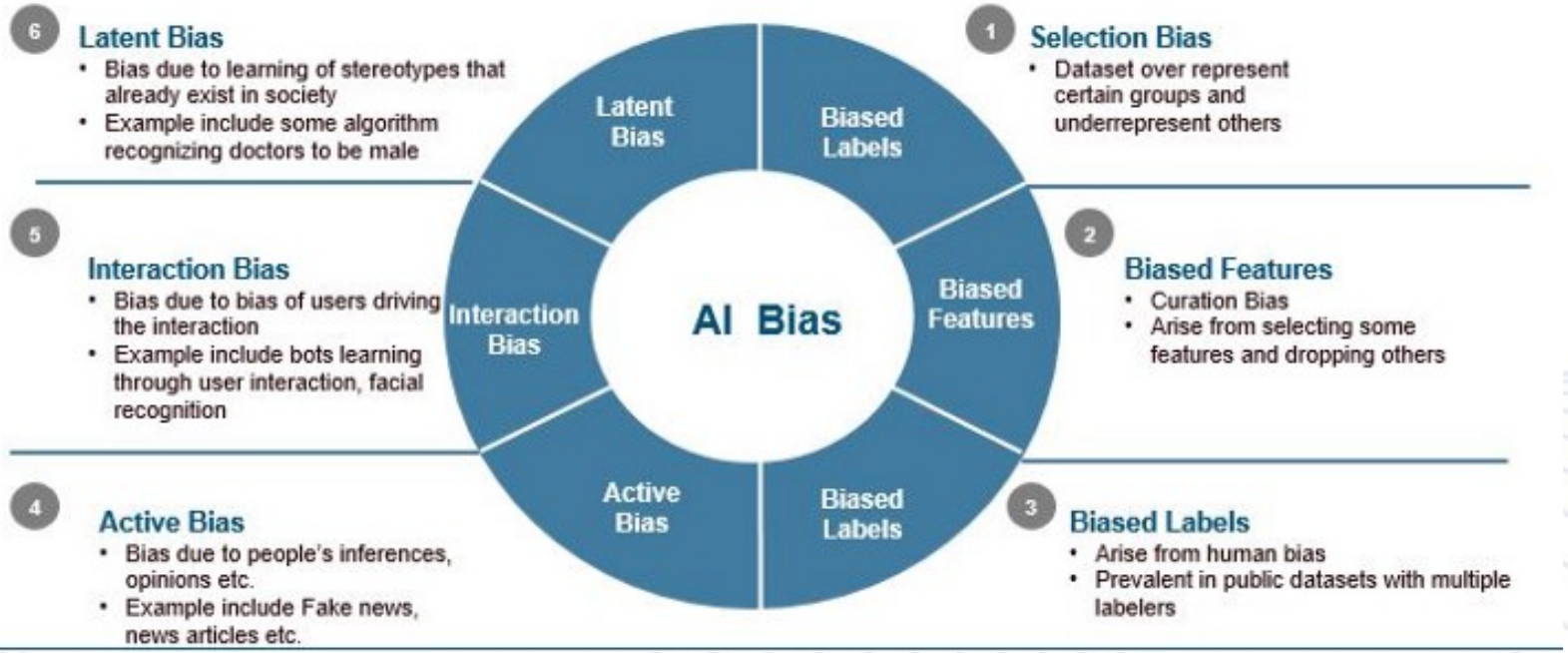
Mark Twain

Introduction



Source: British Medical Journal

Introduction



Source: ichi.pro: "Bias-Erkennung in Modellen fÃ¼r maschinelles Lernen mit Amazon SageMaker Clarify"

Bias

Definition: Influences on the human mind

Types of Bias

1. Gender Bias
2. Color Bias
3. Ageism (Age Discrimination)
4. Bias due to the name
5. Beauty Bias
6. Halo Effect
7. Horns Effect
8. Confirmation Bias
9. Conformity Bias
10. Affinity Bias
11. Contrast effect
12. Status quo setting
13. Anchor effect
14. Authority Bias
15. Overconfidence bias
16. Perception error
17. Bogus correlation
18. Affect heuristics
19. Recency effect
20. Idiosyncratic valuation errors

Types of Bias

Gender Bias

Gender bias, the preference of one gender over another, is often referred to as sexism. This bias occurs when a person unconsciously associates certain stereotypes with different genders.

see: Gender bias in artificial intelligence - Consilium
and

News of the European Commission against Racism and Intolerance (ECRI)

Gender Bias - Example

How AI systems amplify bias

Image recognition systems that use biased machine learning data sets will inadvertently magnify that bias. Researchers are examining ways to reduce the effects.

COOKING

ROLE	VALUE
AGENT	WOMAN
FOOD	PASTA
HEAT	STOVE
TOOL	SPATULA
PLACE	KITCHEN

COOKING

ROLE	VALUE
AGENT	WOMAN
FOOD	FRUIT
HEAT	–
TOOL	KNIFE
PLACE	KITCHEN

COOKING

ROLE	VALUE
AGENT	WOMAN
FOOD	MEAT
HEAT	GRILL
TOOL	TONGS
PLACE	OUTSIDE

COOKING

ROLE	VALUE
AGENT	WOMAN
FOOD	VEGETABLES
HEAT	STOVE
TOOL	TONGS
PLACE	KITCHEN

COOKING

ROLE	VALUE
AGENT	MAN
FOOD	–
HEAT	STOVE
TOOL	SPATULA
PLACE	KITCHEN

In this example of gender bias, adapted from a report published by researchers from the University of Virginia and the University of Washington, a visual semantic role labeling system has learned to identify a person cooking as female, even when the image is male.

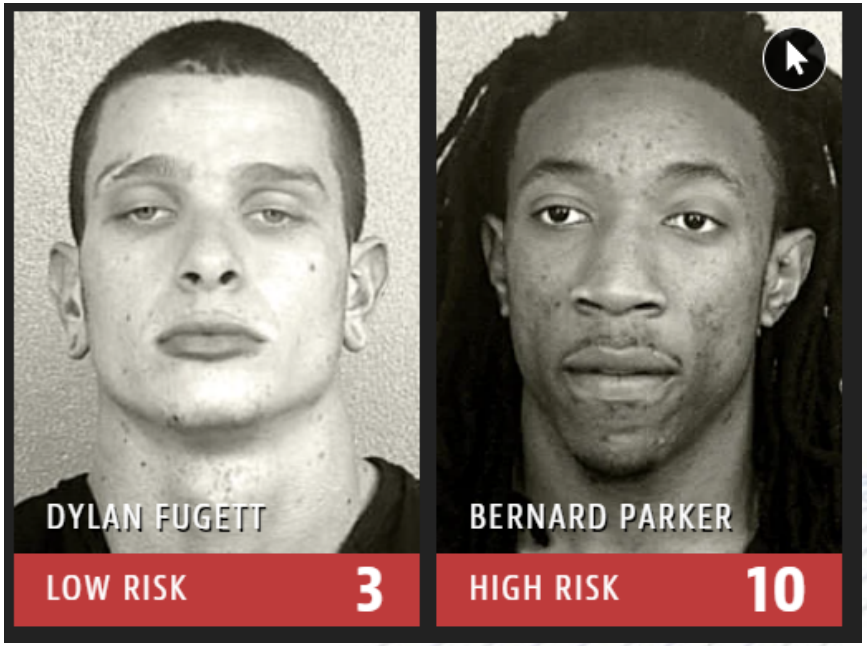
Gender Bias - Avoidance

- Establish gender-neutral hiring standards: Define the ideal candidate profile in advance and evaluate all candidates against these standards.
- Create diversity targets: Set qualitative gender diversity goals to create a more gender-balanced team. Support women in taking on leadership roles and provide resources for this.

Color Bias

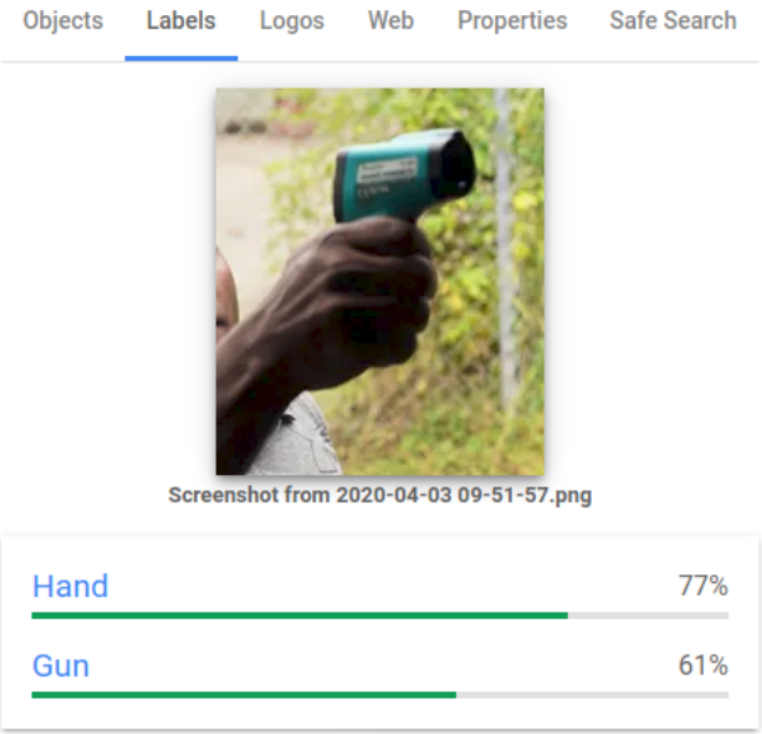
Color bias, the preference of one color over another, is often referred to as racism. This bias occurs when a person unconsciously associates certain stereotypes with different colors.

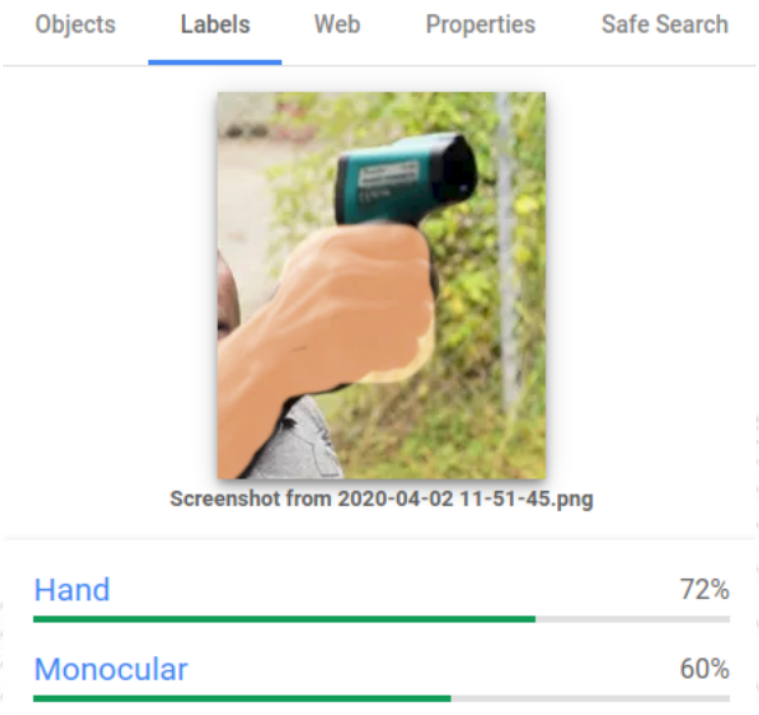
Color Bias - Example



see: Here's software used across the country to predict future criminals. And it's biased against blacks.

Color Bias - Example





see: Google apologizes after its Vision AI produced racist results

Color Bias - Avoidance

- Establish color-neutral hiring standards: Define the ideal candidate profile in advance and evaluate all candidates against these standards.
- Create diversity targets: Set qualitative color diversity goals to create a more color-balanced team.

Ageism

Ageism refers to stereotyping or discriminating against others based on their age, which often occurs with older team members.

Ageism - Example

Ageism - Avoidance

- Don't make assumptions based on age: So don't automatically assume that older employees don't know how to use new technologies or are unwilling to learn new skills. Provide equal training opportunities for all.
- Encourage cross-generational collaboration: Create reciprocal mentoring programs where an older team member works with a new employee. This type of collaboration facilitates communication between team members of different ages, which can help break down misconceptions about age.

Bias due to the Name

Name bias is the tendency to prefer certain names, usually German-sounding names, over others.

Bias due to the Name - Example

Kevin: "No name, a diagnosis"

Source: "Wie es sich anfühlt, ein Kevin zu sein", Welt, 2015

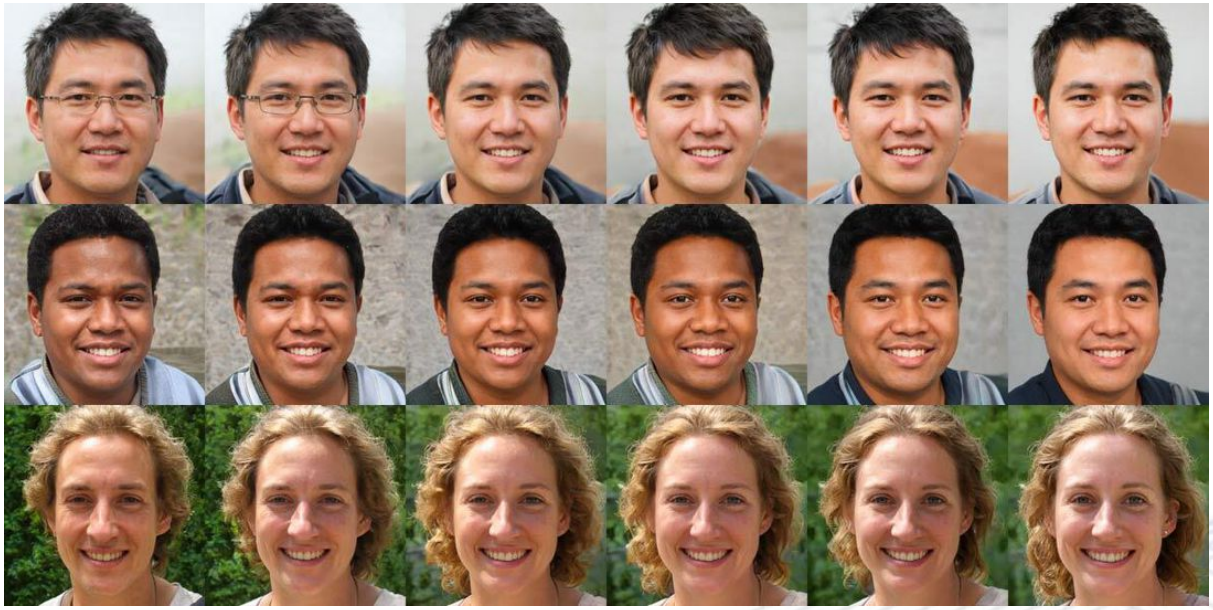
Bias due to the Name - Avoidance

- Use software: Use blind hiring software that hides applicants' personal information on resumes.
- Manual approach: Designate a team member to remove personally identifiable information from resumes for the hiring team.

Beauty Bias

The term “beauty bias” describes the preferential treatment and positive stereotyping of people who are considered more attractive. This has also given rise to the term “lookism”, which describes discrimination based on physical appearance.

Beauty Bias - Example



Researcher Bogdan Kulynych discovered that Twitter's AI algorithm typically favored youthful, fair-skinned, and slimmer versions of an initial image.

see: Twitter AI bias contest shows attractiveness filters hoodwink the algorithm

Beauty Bias - Avoidance

- Omit photos from resumes: When reviewing resumes, focus on the applicant's qualifications and experience.
- Pre-screen by phone: Before scheduling an interview, consider conducting a brief interview over the phone to get to know applicants better without being influenced by their appearance.

Halo Effect

Coined by psychologist Edward Thorndike in the 1920s, the halo effect occurs when we develop an overall positive impression of a person based on a single trait or merely one characteristic.

Halo Effect - Example

Halo Effect - Avoidance

- Conduct multiple interviews: Schedule multiple rounds of interviews for applicants with different levels of management. This allows an applicant to be evaluated from different perspectives.
- Conduct a diverse team for the interview: It can be helpful to bring in a person from another team to interview. This person has less reason to put a "halo" on the applicant because they will not be working directly with the prospective team member.

Horns Effect

The Horns effect is the opposite of the halo effect. This bias causes us to have a negative impression of a person based on a single characteristic or experience.

Horns Effect - Example

Horns Effect - Avoidance

- Challenge first impressions: Take the time to get to know a person so you can develop a more concrete impression of them as a whole.
- Make judgments based on facts: Ask yourself how you developed your first impression of a person and find evidence to support or refute that impression based on further interactions.

Confirmation Bias

Confirmation bias is the tendency to seek and use information that confirms one's own views and expectations. In other words, this form of unconscious bias arises from selecting information that confirms certain assumptions.

Confirmation Bias - Example

Confirmation Bias - Avoidance

- Obtain information from multiple sources: Whenever you are testing a hypothesis or conducting market research, you should gather information from a variety of sources to get as balanced a perspective as possible.
- Standardize interview questions: When selecting new employees, establish a list of standard interview questions to avoid asking off-topic or pointed questions that may or may not confirm your opinion of an applicant.

Conformity Bias

Conformity bias is similar to groupthink, where we conform our opinion or behavior to that of the group, even if it does not match our own opinion.

Conformity Bias - Example

Conformity Bias - Avoidance

- Use anonymous polls or questionnaires: The option to give feedback anonymously allows more freedom to express your opinion without worrying about the preferences of others.
- Ask for opinions in advance: Before going into a meeting, have a one-on-one conversation with each team member to get their opinion. This gives everyone enough time to think about a topic and make their thoughts known without the pressure of having to speak in front of colleagues.

Affinity Bias

Affinity bias, also known as affinity bias or similarity bias, refers to the tendency to prefer people who share interests, ethnicity, sexual orientation, and experiences with us. We tend to feel more comfortable around people who are similar to us.

Affinity Bias - Example

Affinity Bias - Avoidance

- Build a diverse hiring team: Diverse individuals with different perspectives and interests interviewing candidates can help mitigate an individual's affinity bias.
  Pay more attention than just "cultural fit". The more common ground recruiters have with job applicants, the more likely they are to assess them as a "cultural fit". However, the term "cultural fit" is vague and can mean different things to different people. To fairly evaluate applicants, use specific wording and examples when providing feedback about them. Describe how well they embody company values or align with company goals.

Contrast Effect

We often make judgments by making comparisons. Consequently, our judgments can change depending on the standard to which we compare something: This is called the contrast effect.

Contrast Effect - Example

Contrast Effect - Avoidance

- Make multiple comparisons: Instead of drawing a conclusion after just one comparison, compare a fact to different standards to broaden your perspective.
- Provide explanations: Explain to your colleagues how you came to a particular conclusion so they can understand your point of view.

Status Quo Setting

This type of bias describes a preference for the status quo, or for keeping things the way they are, which can lead to resistance to change.

Status Quo Setting - Example

Status Quo Setting - Avoidance

- Leverage the framing effect: We often stay with the status quo to avoid a loss, which we weight more heavily compared to a benefit. The framing effect involves viewing the default option as a loss to encourage exploration of alternative options as a gain.
- Thinking outside the box: Create an environment that rewards creativity and innovation. Enable an open attitude toward change so your team can continue to push the status quo.

Anchor Effect

An anchor effect occurs when we rely too much on the first piece of information we receive as an anchor and base our decisions on it. In doing so, we look at the facts from a narrow perspective.

Anchor Effect - Example

Anchor Effect - Avoidance

- Research thoroughly: The first option may not always be the best. Look into different possible options and their advantages and disadvantages before making a final decision.
- Brainstorm with the team: Discussing a particular decision with teammates can help reveal the strengths and weaknesses of a plan.

Authority Bias

The term "authority bias" refers to the tendency to believe in authority figures and to follow their instructions.

E.g.: Person with experienced, algorithms, documentation, google

Authority Bias - Example

Authority Bias - Avoidance

- Ask questions: Don't be afraid to ask your supervisor or the company manager questions. The level of detail he provides can be an indication of whether an idea was well thought out or if this is just his authority coming into play.
- Conduct research: Conduct your own research on a particular topic to identify other credible sources or experts and see if their suggestions match those of your supervisor.

Overconfidence Bias

Unconscious bias due to overconfidence, also known as overconfidence bias, arises from people's tendency to think they are better at certain skills and abilities than they actually are.

Overconfidence Bias - Example

Overconfidence Bias - Avoidance

- Consider consequences: The decisions you make can impact your business. Before making a decision, identify all possible consequences to ensure you are prepared.
- Ask for feedback: Soliciting feedback from your team will help you identify opportunities for improvement, whether it's your performance or your ideas. Constructive criticism can keep an ego in check.

Perception Error

Perceptual biases occur when we judge or treat other people based on often inaccurate, oversimplified stereotypes and preconceptions about the social groups to which they belong. Other biases such as gender, age, and appearance may also play a role.

Perception Error - Example

Perception Error - Avoidance

- Challenge preconceptions: ask yourself, "How well do I really know this person or the social group they belong to?" Don't let a preconceived notion keep you from meeting or involving new people.
  Reflect on the accuracy of statements: when you find yourself using strong words like "all", "always", and "never" to describe a particular group, pause and take a moment to ask yourself how accurate the description really is.

Bogus Correlation

A spurious correlation is when we associate two variables, events, or actions with each other even though they are unrelated.

Bogus Correlation - Example

Bogus Correlation - Avoidance

- Educate yourself: Learning more about the areas you are not familiar with can help you find evidence that supports or refutes the correlation.
- Consider all possibilities: When relating two facts, consider the probability of cause and effect. You can also use a contingency table to illustrate the relationship between cause and effect.

Affect Heuristics

Heuristics are mental shortcuts that help us make more efficient and better decisions. The emergence of this form of Unconscious Bias is due to the fact that we rely on our emotions when making decisions. This allows us to reach a conclusion more quickly, but it is not always correct or fair.

Affect Heuristics - Example

Affect Heuristics - Avoidance

- Be aware of emotions: Being aware of your emotions in a situation allows you to gain distance from the situation and evaluate it more logically.
- Take time to reflect: Reflect on an event some time after it has occurred. Your emotions will probably not be as strong as they were during the event, which will enable you to come to a more objective conclusion.

Recency Effect

A recency effect occurs when we attach greater importance to current events than to past ones because we can remember them more easily.

Recency Effect - Example

Recency Effect - Avoidance

- Take notes: Take detailed notes during each interview, and review them afterwards. This will help you keep track of noteworthy applicants, regardless of when you had the conversation with them.
- Take breaks: Conducting multiple interviews in a row can be mentally taxing. When your working memory takes its toll, you're more likely to be affected by the recency effect. Stay mentally alert by taking a break between each job interview so your brain has time to absorb and retain the information.

Idiosyncratic Valuation Errors

An idiosyncratic evaluation bias affects the way we evaluate the performance of others. We often evaluate others based on our subjective interpretations of the evaluation criteria and our own definition of what "success" looks like.

Idiosyncratic Valuation Errors - Example

In other words, we are generally unreliable when it comes to evaluating others. Research has shown that about 60% of a supervisor's evaluation reflects the supervisor, not the team member he or she is evaluating.

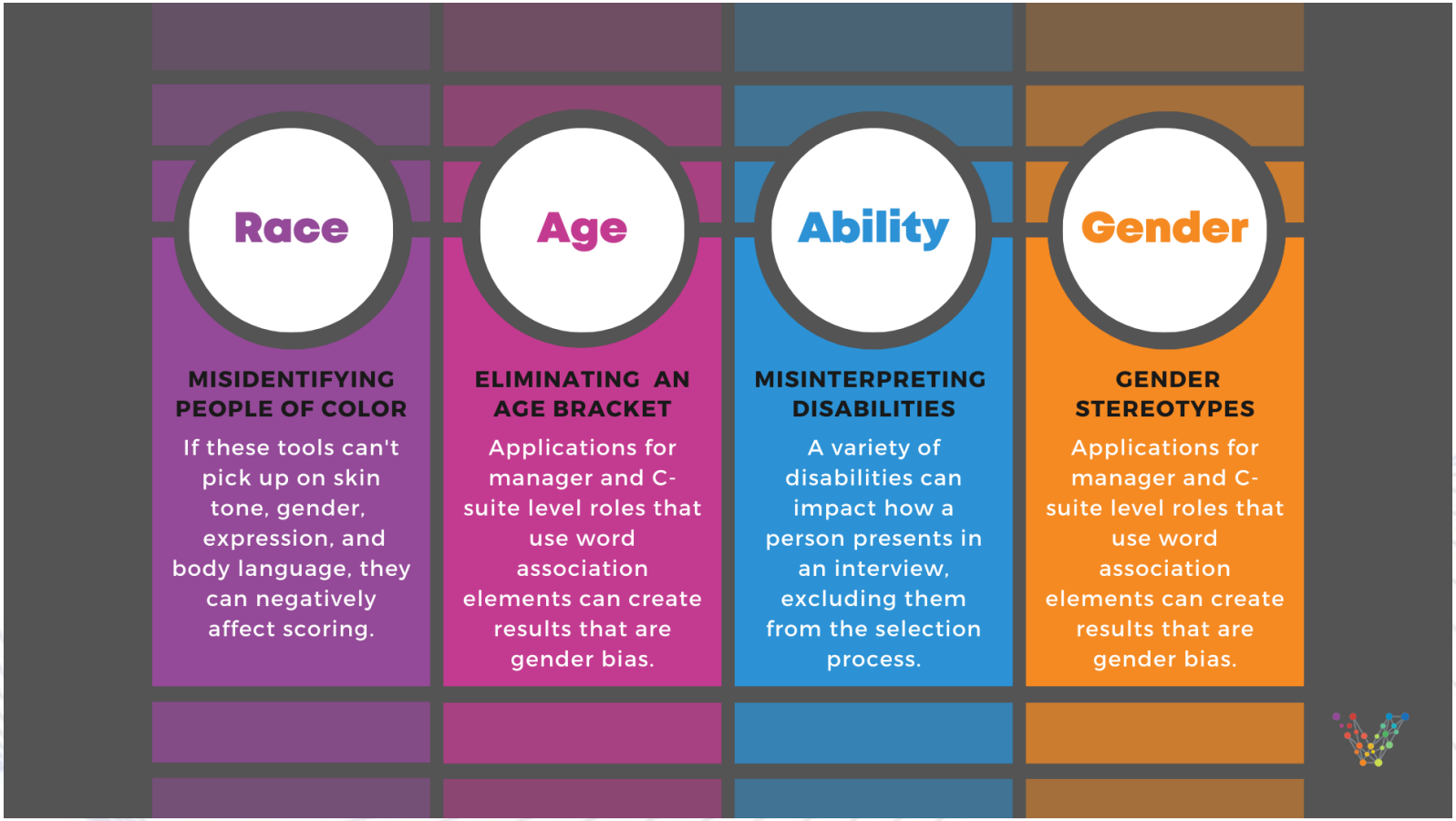
For example, a supervisor who excels at project management has higher standards for this skill and gives team members more stringent ratings for it. On the other hand, the supervisor is more lenient in assessing the marketing skills of his team members because he is less familiar with this area.

Idiosyncratic Valuation Errors - Avoidance

- Establish concrete and clear evaluation criteria: create a rubric or concrete set of standards for evaluating performance. This will prompt supervisors to provide evidence based on a team member's performance or accomplishments to determine how well they did.
- Conduct multi-rater evaluations: In this process, a team member receives feedback from peers and supervisors in addition to their self-assessment. Multiple ratings can help supervisors get a more comprehensive view of a team member's performance and identify potential areas for growth.

Examples

Example - The New HR



see: Tyler Rouse: The New HR: How AI Evolved the Hiring Process